a

b

Congenital phenotypes

c

Non−congenital phenotypes

d

HPO phenotype

Atrophy/Degeneration affecting the central nervous system
Anencephaly
Hydranencephaly
Fetal akinesia sequence
Arrhinencephaly
Iniencephaly
Congenital neuroblastoma
Neonatal seizure with bilateral symmetric automatism
Diastomatomyelia
Absent mesencephalon
profound

Posterior predominant thick cortex pachygyria
Hypocapnia in cord blood
Abnormal CSF carnosine concentration
Decreased circulating alpha−mannosidase activity
Pulmonary xanthomatosis
Reduced leukocyte cathepsin D activity
Flexner−Wintersteiner rosette
Frontal encephalocele
Morphological central nervous system abnormality
Optic nerve hypoplasia
severe

Abnormal diffusion weighted cerebral MRI morphology
Dysplasia of second lumbar vertebra
Osteolytic defects of the 1st metacarpal
Abnormality of the peripheral nervous system
Retinitis pigmentosa inversa
Symmetrical progressive peripheral demyelination
Vestibular saccular degeneration
Abnormal peripheral nervous system morphology
Chiari malformation
Tethered cord
moderate

Abnormal circulating glucose−6−phosphate dehydrogenase concentration
Abnormal glucose−6−phosphate dehydrogenase level in blood
Abnormal size of nasopharyngeal adenoids
Abnormality of the scalp hair
Atrichia
Elevated urinary aminoisobutyric acid
Fish odor
Hypoornithinemia
Overly formal/pedantic language
Thick hair
mild

Brainstem glioma
Pineal parenchymal cell neoplasm
Atrophy/Degeneration affecting the cerebrum
Chronic hepatic encephalopathy
Acute necrotizing encephalopathy
Astrocytoma
Cerebellar medulloblastoma
Ependymoblastoma
Vegetative state
Abnormal corticospinal tract morphology
profound

Alveolar soft part sarcoma
Carcinoma
Orbital rhabdomyosarcoma
Testicular teratoma
Acute myelomonocytic leukemia
Localized neuroblastoma
Acute lymphoblastic leukemia
Anaplastic thyroid carcinoma
Neoplasm by anatomical site
Disseminated viral infection
severe

Pituitary acidophilic stem cell adenoma
Abnormal upper motor neuron morphology
Anti−Sj/ITPR1 antibody
Cranial nerve compression
Systemic lupus erythematosus
Anti−H2A antibody positivity
Anti−H3 antibody positivity
Anti−histone antibody positivity
Anti−myeloperoxidase antibody positivity
Pituitary spindle cell oncocytoma
moderate

Abnormal psychotic pattern
Addictive video game use
Compulsive stealing
Overvalued idea
Prolonged N2 sleep
Prolonged N3 sleep
Prolonged rem sleep
Religious delusion
Short rem sleep
Sleep onset rapid eye movement period
mild

death
intellectual disability
impaired mobility
blindness
physical malformations
sensory impairments
immunodeficiency
cancer
reduced fertility
congenital onset

Occurrence
never
rarely
often
always

Severity score
40
30
20
10
0